Abstract

The transition from a linear “take-make-dispose” model to a Circular Economy (CE) presents a significant paradigm shift in corporate strategy, particularly within asset heavy industries such as waste management. While a large body of literature has explored the conceptual benefits of CE, a critical gap remains in empirical research on its direct financial impact, especially when considering the role of tangible strategic assets. This dissertation addresses this gap by investigating the distinct and combined effects of a quantifiable CE strategy (recycling revenue) and a tangible strategic asset (landfill ownership) on long-term shareholder value.

Adopting a comparative case study design, this research examines three publicly traded firms: Waste Management Inc., Republic Services, and Waste Connections. The methodology employs a common-size financial analysis of key metrics, including Return on Equity, Enterprise Value to EBITDA, and Total Shareholder Return, spanning an eight-year period from 2017 to 2024. The analysis is guided by the Resource-Based View (RBV) and Transaction Cost Economics (TCE) to understand how the control of valuable, rare, and difficult-to-imitate assets creates a durable competitive advantage. Signalling Theory is also relevant in the context of voluntary sustainability initiatives, as these actions can serve as credible signals of a firm's long-term viability and ethical commitment.

The findings reveal that landfill ownership is a more significant determinant of long-term shareholder value than CE initiatives alone. Landfill assets, characterized by high barriers to entry and vertical integration, provide a stable foundation for revenue and strategic flexibility, directly translating into premium market valuations. In contrast, while CE strategies are crucial for social license and risk management, acting as a positive signal to the market, their financial returns in this industry appear to be a confounding factor rather than a direct driver of superior shareholder value. The study concludes that for firms in asset heavy sectors, a holistic approach to value creation must balance socially driven CE initiatives with the control of foundational, revenue generating strategic assets to achieve sustainable, long-term returns.

Table of contents

Chapter 1: Introduction 4

1.1 Background and Context 4

1.2 Research Problem & Gap 4

1.3 Research Question 4

1.4 Objectives 5

1.5 Dissertation Structure 5

Chapter 2: Literature Review 6

2.1 Circular Economy, ESG, and Value Creation 6

2.2 Strategic Assets and Foundational Infrastructure 6

2.3 Theoretical Integration and Conceptual Framework 7

Chapter 3: Methodology 9

3.1 Research Design 9

3.2 Case Selection 9

3.3 Data Collection 9

3.4 Metrics and Variables 10

3.5 Analytical Tools and Limitations 10

Chapter 4: Analysis 11

4.1 Common-Size Financial Analysis: Operational and Capital 11

4.2 Common-Size Balance Sheet and Capital Structure Analysis 13

4.3 Strategic Assets, Circular Economy, and Shareholder Value 14

4.4 Conclusion of Analysis 16

Chapter 5: Discussion 17

5.1 Landfill Ownership as a Differentiated Strategic Asset 17

5.2 Strategic Agility and Premium Market Valuation 17

5.3 The Confounding Role of Circular Economy Strategies 18

5.4 Summary of Discussion 19

Chapter 6: Conclusion 20

6.1 Summary of Findings 20

6.2 Answering the Research Question and Contributions 20

6.3 Final Thoughts, Limitations, and Future Research 21

6.4 Broader Implications: Beyond Shareholder Value 21

Chapter 7: Appendix 22

7.1 Waste Management Inc. *22*

7.2 Waste Connections *26*

7.3 Republic Services *29*

7.4 Combined Data *33*

Chapter 8: References 37

Chapter 1: Introduction

1.1 Background and Context

Corporate operations are undergoing a paradigm shift, moving from a linear “take-make-dispose” model to a Circular Economy (CE) and embracing Environmental, Social, and Governance (ESG) principles (Ghisellini, Cialani & Ulgiati, 2016; Geissdoerfer et al., 2017). The Circular Economy, a strategic response, promotes the continuous use of resources through reuse, recycling, and waste-to-energy conversion (Bocken et al., 2016). For the waste management sector, this shift is fundamental, as it is uniquely positioned to transition from a disposal centric to a resource recovery model. This transition requires significant capital investment in infrastructure, which leads to a central element of this study: landfill ownership. Landfills are increasingly recognized as strategic assets. Their value lies in barriers to entry, such as site scarcity and complex permitting processes. Owning a landfill provides a company with a captive disposal point, enabling vertical integration and long-term pricing power (Linder & Williander, 2015). This control offers a durable competitive "moat." Therefore, this study proposes that both the adoption of CE strategies and the control of landfill assets are determinants of a firm’s long-term financial health and shareholder value.

1.2 Research Problem & Gap

Despite a surge in interest in ESG and the Circular Economy, a critical gap exists in empirical research on their direct, financial impact in asset heavy sectors. Much of the existing CE literature is conceptual, lacking concrete evidence of its financial returns at the firm level (Pomponi & Moncaster, 2016). Similarly, while ESG performance is often linked to positive financial outcomes, studies frequently rely on aggregated scores that fail to differentiate between a company's specific environmental strategies. The value creating mechanisms of tangible environmental assets, such as landfills, are often overlooked. While the Resource-Based View (RBV) focuses on firm value creation (Barney, 2007), it tends to emphasize intangible assets like brand reputation. The financial contribution of physical assets like landfills has been undertheorized. This dissertation contributes to existing literature by empirically investigating the distinct and combined effects of a tangible strategic asset (landfills) and a quantifiable CE strategy (recycling revenue) on shareholder value, offering a more granular analysis than previous studies on broad ESG performance.

1.3 Research Question

"Do circular economy strategies and landfill ownership contribute to long-term shareholder value in global waste management firms?"

1.4 Objectives

This dissertation aims to address the research gap by empirically investigating the relationship between circular economy strategies, landfill ownership, and shareholder value. To achieve this, the following specific objectives will be met:

1. To investigate the specific circular economy strategies and landfill ownership models adopted by key players in the global waste management industry.
2. To compare the impact of these strategies and assets on three key financial metrics: Return on Equity, Enterprise Value to EBITDA, and Total Shareholder Return.
3. To evaluate whether the combination of CE strategies and landfill ownership creates a sustainable competitive advantage that translates into superior shareholder value.

1.5 Dissertation Structure

This dissertation is structured to provide a comprehensive analysis of the research question. The introduction provides the background, context, and a clear statement of the research problem, question, and objectives. The literature review synthesizes existing academic work to establish a theoretical foundation and conceptual framework. The methodology chapter outlines the research design and analytical tools. The analysis chapter presents the empirical findings from the comparative case study. The discussion interprets these findings in relation to the conceptual framework and existing literature. Finally, the conclusion summarizes key findings, discusses the study's limitations, and suggests avenues for future research.

Chapter 2: Literature Review

This chapter provides a literature review on the relationship between circular economy (CE) strategies, landfill ownership, and long-term shareholder value in the waste management sector. It establishes theoretical foundations from CE, Environmental, Social, and Governance (ESG) literature, and the Resource-Based View (RBV). The review then synthesizes debates on how CE practices and asset ownership influence firm competitiveness and financial outcomes, before proposing a conceptual framework to guide the research.

2.1 Circular Economy, ESG, and Value Creation

The circular economy is a strategic shift from the traditional "take-make-dispose" model, promoting the continuous use of resources through practices like reuse and recycling (Geissdoerfer et al., 2017). Scholars like Ghisellini, Cialani, and Ulgiati (2016) define the CE as a model designed to eliminate waste and generate value by creating closed loop systems (Bocken et al., 2016). While CE is widely associated with benefits such as innovation and a strong reputation (Lacy & Rutqvist, 2015), empirical evidence directly linking CE to financial outcomes remains limited, especially in asset heavy industries where substantial infrastructure is required for implementation (Pomponi & Moncaster, 2016).

To understand the financial implications of CE, it is essential to examine the ESG framework, which generally finds a positive link between strong ESG performance and financial outcomes, including higher stock returns and a lower cost of capital (Friede, Busch & Bassen, 2015; Clark, Feiner & 2015). These financial benefits are often driven by increased stakeholder trust, improved reputation, and better risk management (Kotsantonis, Pinney & Serafeim, 2016). However, the broad nature of aggregated ESG scores can make it difficult to isolate the financial impact of specific environmental strategies like CE. This is why materiality is key: as argued by Khan, Serafeim, and Yoon (2016), only financially relevant ESG issues truly influence performance. In the waste sector, CE practices and waste handling assets are directly tied to costs and revenue, making them highly material to the business. This study acknowledges that CE encompasses a range of practices, including innovative product design and supply chain partnerships, but will focus on revenue generating activities such as recycling and energy recovery which are directly measurable for financial analysis.

2.2 Strategic Assets and Foundational Infrastructure

The Resource-Based View (RBV) (Barney, 2007; Wernerfelt, 1984) explains that long-term competitive advantage comes from controlling resources that are valuable, rare, hard to copy, and non-substitutable (VRIN). In waste management, assets such as landfills, recycling centres, and energy recovery facilities can function as these strategic resources. Implementing a large scale CE model is heavily dependent on this physical infrastructure (Pomponi & Moncaster, 2016). However,

a significant portion of the CE literature overlooks the importance of a firm's ownership of this key infrastructure, focusing instead on abstract models and business practices (Korhonen et al., 2018).

Landfill ownership, in particular, offers significant advantages by creating barriers to market entry, enabling vertical integration, and providing long-term pricing power (Williamson, 1985). These assets meet the VRIN criteria because of site scarcity and the complex, regulated permitting processes (Williamson, 1985). In the waste sector, controlling landfills offers a unique position for firms to manage waste streams and monetize recovered materials, enhancing performance and investor appeal. Despite this clear link, few academic studies have empirically tested this specific relationship within waste firms, highlighting a gap this research aims to address.

2.3 Theoretical Integration and Conceptual Framework

To explain the relationship between CE strategies, landfill ownership, and shareholder value, this study presents a conceptual framework (Figure 1) that integrates the Resource-Based View (RBV) with Materiality and Signaling Theory.

According to the RBV, a firm's inputs, landfill ownership and CE initiatives are key strategic assets. Landfills meet the VRIN criteria due to site scarcity and complex regulation (Barney, 2007; Williamson, 1985). The framework explains that these strategic inputs influence a company's financial profile by affecting its capital structure and operational efficiency. Implementing CE initiatives, for example, improves resource efficiency and reduces disposal costs (Pomponi & Moncaster, 2016).

These operational and structural changes, in turn, are expected to influence a firm’s financial performance and market valuation. Research shows that sustainability and CE strategies can lead to higher valuations through cost savings, diversified revenue, and risk mitigation (Linder et al., 2017). This is further supported by Materiality Theory which suggests that investors focus on issues directly impacting a company's financial profile (Khan, Serafeim & Yoon, 2016). Strong performance on these material issues, such as landfill control and recycling revenue can also act as a positive signal to investors, a concept rooted in Signaling Theory (Spence, 1973). This signals a firm's forward looking strategic vision and stability, which can contribute to a premium market valuation. The entire framework provides a way to test if these specific strategies and assets lead to long-term shareholder value.

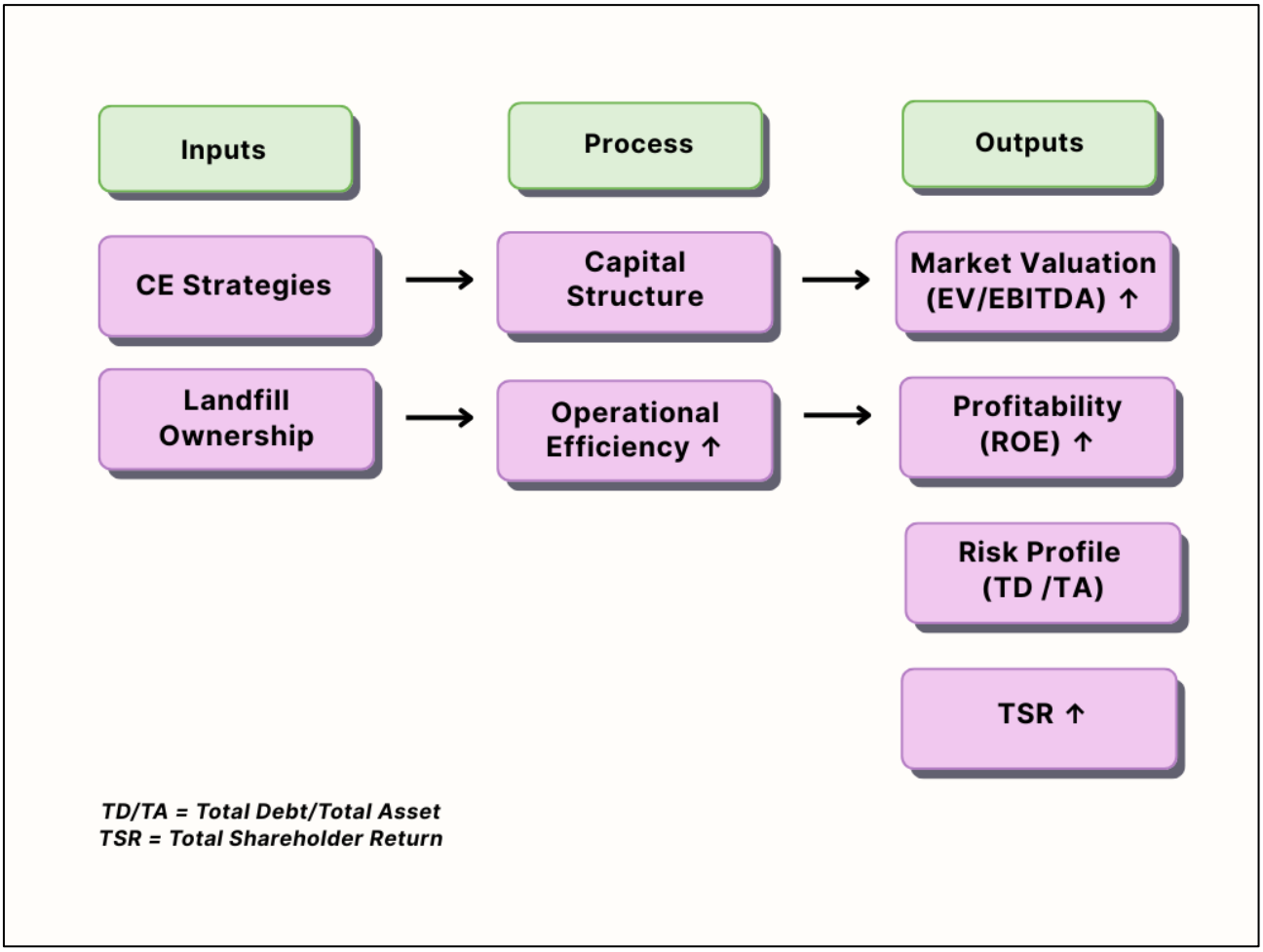


Figure 1: Linking CE Strategy and Landfill Ownership

Chapter 3: Methodology

This chapter outlines the research methodology used to investigate how circular economy (CE) strategies and landfill ownership contribute to long-term shareholder value in the North American waste management sector. It details the research design, justifies the selection of cases, explains the data collection methods, and defines the variables and analytical tools used. The rigorous approach taken ensures the reliability and validity of the findings in addressing the central research question.

3.1 Research Design

This study adopts a comparative case study approach, which is appropriate for exploring a complex, contemporary phenomenon within its real world context (Yin, 2018). This design allows for an in depth exploration that a purely quantitative study could not provide, given the nuanced interplay between CE strategies, unique assets, and financial performance. By comparing multiple cases, the research aims to identify commonalities and patterns. A central tenet is the triangulation of data, which enhances credibility (Creswell, 2014). While a descriptive design has limitations in establishing causal relationships, it is a suitable method for an exploratory, Master's level dissertation, providing new insights and a foundation for future research. Qualitative data from corporate reports is combined with quantitative financial data and third party sustainability ratings to provide a comprehensive perspective.

3.2 Case Selection

The study focuses on three prominent North American waste management firms: Waste Management Inc., Republic Services, and Waste Connections. This deliberate selection is a purposeful sampling choice, justified by several key criteria aligned with the research question. These firms are among the largest and most influential market leaders with extensive landfill infrastructure, making them ideal subjects to assess the strategic implications of such assets. Each company has publicly implemented various CE strategies, allowing for an examination of their strategic commitment. Finally, their status as publicly traded companies ensures the availability of robust financial data for assessing shareholder value.

3.3 Data Collection

This dissertation relies exclusively on secondary research, a method well suited for analysing historical and publicly disclosed strategic information (Creswell & Creswell, 2018). This approach ensures consistency and allows for analysis of trends over the 2017-2024 period. Financial data is primarily sourced from firms' annual reports (10−K filings) and investor presentations. This approach aligns with Signaling Theory (Spence, 1973), as it uses the same information available to investors and analysts to form their valuation decisions, providing a robust basis for empirical

analysis. This data is cross referenced with reputable financial databases like Bloomberg, to ensure accuracy. Qualitative and quantitative data on CE initiatives and landfill assets are extracted from the companies' corporate sustainability and annual reports.

3.4 Metrics and Variables

The independent variables are Circular Economy (CE) Strategy and Landfill Ownership. CE strategy is measured through a qualitative assessment of reported initiatives and quantitative proxies like revenue from recycled materials. Landfill ownership is measured by the number of owned landfills and its revenue contribution. The dependent variables are measures of Shareholder Value and Financial Performance, including Total Shareholder Return (TSR), Return on Equity (ROE), and Enterprise Value to EBITDA (EV/EBITDA). This comprehensive set of variables allows for a robust analysis of how the independent variables may influence financial outcomes.

3.5 Analytical Tools and Limitations

The analysis will primarily use ratio and trend analysis, including common-size analysis to compare firms of different sizes. Financial tables and charts will illustrate trends and highlight disparities. The aim is to observe potential correlations and patterns, rather than to infer statistical causation. It is important to acknowledge the limitations of this methodology. The reliance on public secondary data means the analysis may lack granular detail and carries the inherent bias of corporate self-reporting. The small sample size limits the generalizability of findings, and the study cannot definitively establish a causal link between CE/landfill ownership and shareholder value. The lack of uniform data reporting across firms also requires careful interpretation.

Chapter 4: Analysis

This chapter provides a comprehensive comparative analysis of Waste Management, Republic Services, and Waste Connections from 2017 to 2024. Using common-size analysis to normalize for scale, it investigates how a firm’s circular economy (CE) strategies and landfill ownership affect shareholder value. Landfill ownership is explained by high depreciation, depletion, and amortization (DD&A) as a percentage of revenue, while CE strategies are measured by the percentage of revenue from recycling and energy initiatives. The impact on shareholder value is analysed through key metrics: Return on Equity (ROE), the Enterprise Value to EBITDA (EV/EBITDA) ratio, and Total Shareholder Return (TSR). This quantitative foundation is crucial to the dissertation's argument that strategic assets and forward looking initiatives are key drivers of value.

4.1 Common-Size Financial Analysis: Operational and Capital

A common-size analysis of the income statements reveals distinct operational profiles. By expressing each expense as a percentage of total operating revenues, it becomes evident where each firm allocates its resources and generates profitability. As the two largest firms, Waste Management and Republic Services consistently demonstrate high profitability. Their operating profit (EBIT) margin has remained strong, generally ranging between 17% and 19%. For example, in 2024, Waste Management's EBIT was 18.42% of its revenue, while Republic Services' was 19.94% (Appendix, Table 7.4.4). This consistency suggests robust pricing power and effective cost management, hallmarks of companies with significant market share and scale. This operational stability is a direct result of their vertically integrated business models, which provide a predictable and resilient revenue base, insulating them from the volatility that might affect smaller players.

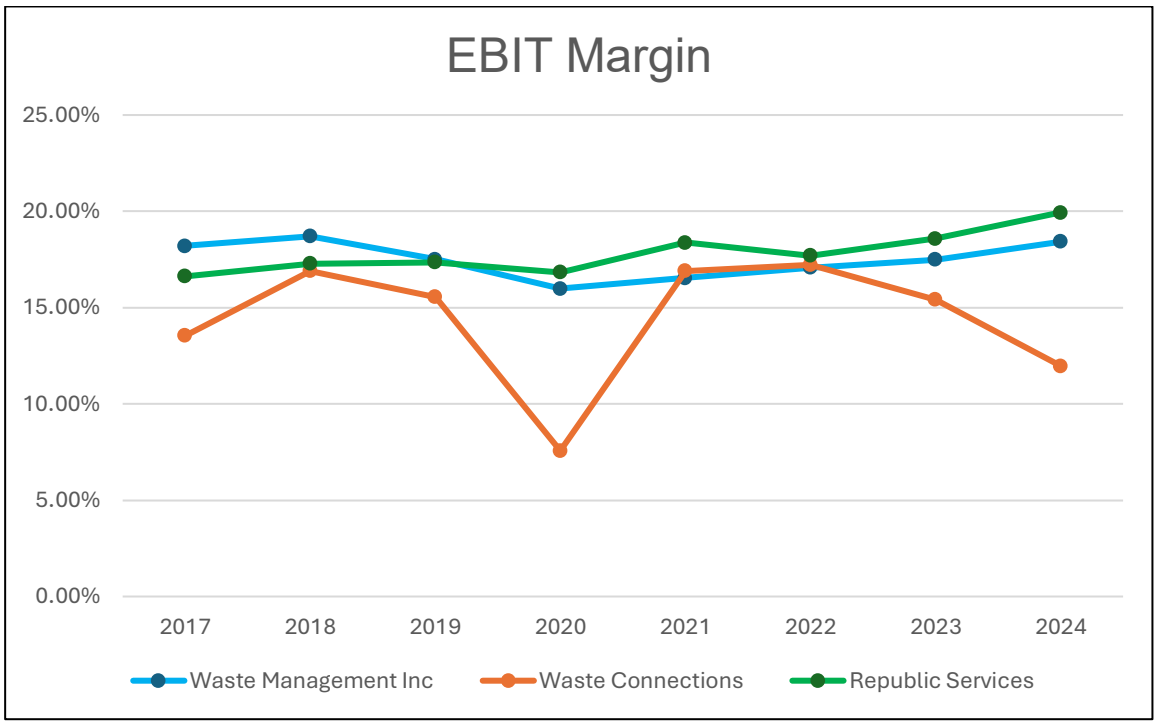


Figure 2: Common-Size Operating Profit

In contrast, Waste Connections exhibits a different operational profile. Its EBIT as a percentage of revenue has fluctuated but has settled in a range similar to its larger peers. In 2024, its EBIT percentage was 11.97%, lower than Waste Management's and Republic Services' for the same year. This difference is a direct function of its targeted growth strategy, which company reports confirm is based on pursuing a greater number of smaller acquisitions. These deals, while fuelling growth, can introduce short term integration costs that impact the common-size profitability metrics more significantly than the larger, less frequent acquisitions of its peers. The analysis of these common-size metrics provides strong evidence that a firm's operational strategy, whether focused on scale or agility, directly influences its profitability profile, supporting the core argument that different strategic asset bases lead to different financial outcomes.

The most revealing metric for this study is depreciation, depletion, and amortization (DD&A) expense as a percentage of revenue. This line item directly reflects a company's capital intensity and its investment in long lived physical assets like landfills. The term "depletion" is particularly relevant here, as it is an accounting method used to expense the consumption of a non-renewable natural resource, in this case, the airspace of a landfill. Thus, a company's DD&A/Revenue ratio is a powerful proxy for its landfill intensity (Hitt, Ireland and Hoskisson, 2013). For Waste Management, DD&A as a percentage of revenue has been consistently high, ranging from 9.50% in 2017 to 10.28% in 2024. Republic Services' figures are similarly high, from 10.32% in 2017 to 10.46% in 2024 (Appendix, Table 7.4.5). These percentages confirm the asset heavy nature of their business models, which are deeply rooted in the ownership of landfills. This high DD&A figure serves as a financial representation of the multi-million dollar investments in landfill expansion and

maintenance that the financial reports highlight, confirming that their operational strategies are heavily dependent on these strategic assets. This finding is central to the dissertation's argument that Resource-Based View (RBV) theory holds true in this industry; the ownership and control of these rare, strategic landfill assets grant WM and RSG a clear competitive advantage that is reflected in their capital structures and profitability.

4.2 Common-Size Balance Sheet and Capital Structure Analysis

Building on the analysis of capital intensity, we now turn to each firm's capital structure, which further explains their strategic choices. The data for the total debt-to-total asset ratio offers a direct quantitative comparison of how each firm leverages debt. Waste Management's total debt-to-asset ratio has seen a significant increase from 43.48% in 2017 to 53.63% in 2024 (Appendix, Table 7.4.6). This rise in leverage is a direct consequence of the company's strategy to fund major, capital intensive projects, most notably the large scale acquisition of US Ecology, Inc. as documented in the financial reports. Republic Services' total debt-to-asset ratio remained relatively stable from 2017 to 2024, reflecting its reliance on debt for asset heavy operations. Meanwhile, Waste Connections' ratio, though starting from a much lower position, rose more aggressively to reach a similar level by 2024. This aligns with its more distributed, less capital intensive strategy of focusing on numerous smaller acquisitions and organic growth, which provides greater agility and a more conservative balance sheet, ultimately leading to lower financial risk and greater fiscal flexibility, a key component of its "asset light" philosophy.

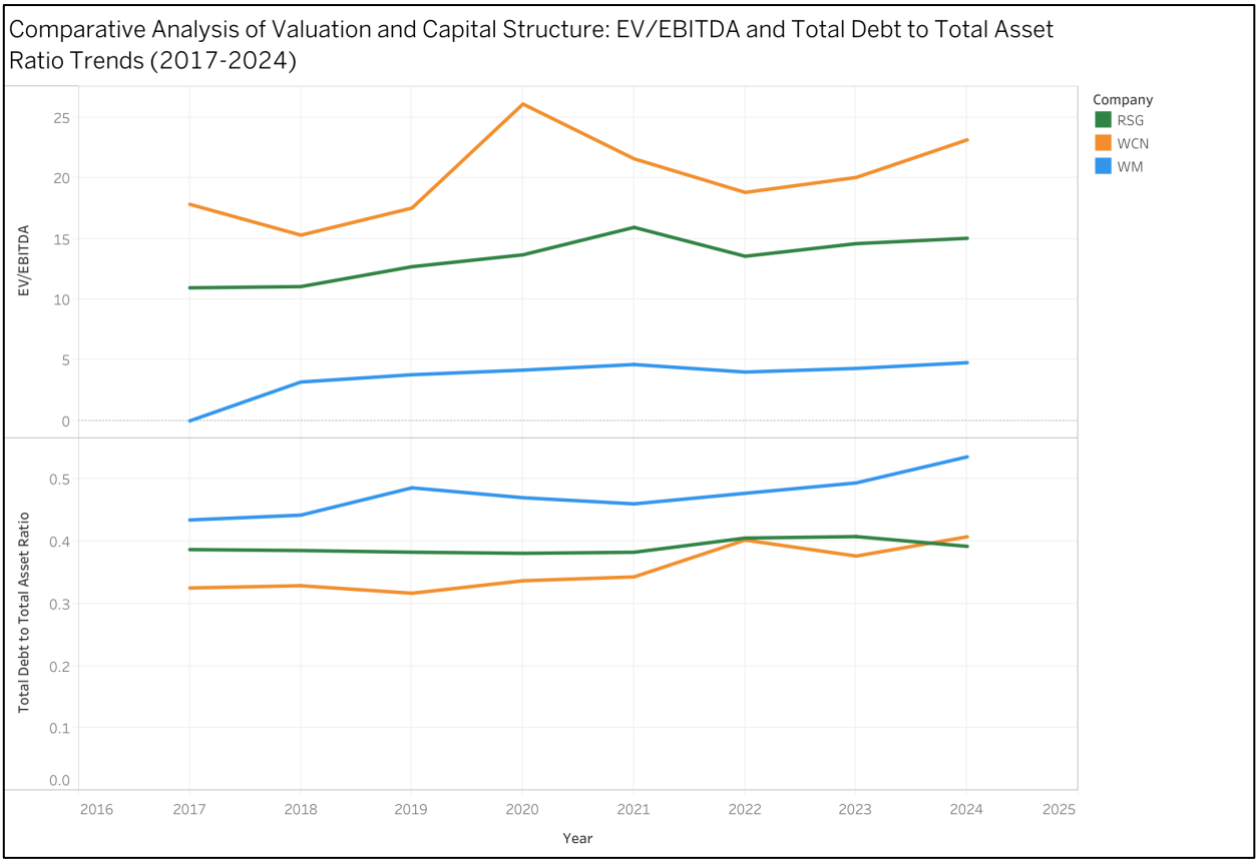


Figure 3: Comparative Trends in Valuation and Capital Structure

The data on landfill investment further highlights these strategic differences. Republic Services' net investment in landfills and the consistent rise in its depletion and amortization expenses directly correlate with its strategy of expanding and optimizing its physical landfill network. These assets are central to RSG's value creation, and the associated costs are a significant factor in its financial performance. This is in sharp contrast to WCN, whose emphasis is less on asset ownership and more on generating high margins from its core service offerings. These capital structure dynamics confirm that a firm's strategic choices regarding asset ownership directly shape its balance sheet, a key component of its financial and operational risk profile.

4.3 Strategic Assets, Circular Economy, and Shareholder Value

Having analysed the foundational financial metrics, we now examine how these strategic differences in asset ownership and CE initiatives translate into shareholder value. The market assigns a different value to each firm's business model, most evident in the Enterprise Value to EBITDA (EV/EBITDA) ratio. Throughout the study period, Waste Connections consistently commands a premium valuation compared to its two larger peers. In 2024, Waste Connections' EV/EBITDA ratio was 23.15, significantly higher than Republic Services at 15.04 and Waste Management at 4.79 (Appendix, Table 7.4.7). This premium valuation is a direct result of the market rewarding its growth oriented model. Investors are willing to pay more for each dollar of WCN's earnings, as they are confident in its future growth potential, operational flexibility, and asset light strategy, perceiving it as a high growth company with a more agile business model.

The data on revenue derived from CE initiatives, provides a critical quantitative measure of each company's strategic focus. The data reveals a clear divergence. In 2024, Republic Services derived an impressive 11.89% of its revenue from circular economy activities, and Waste Management also had a significant portion at 7.27%. This clearly demonstrates that while both WM and RSG have made significant investments in circular economy solutions, WCN has not, as it only generated 2.71% of its revenue from CE initiatives in 2024 (Appendix, Table 7.4.3). This supports the argument that WM and RSG are evolving beyond traditional disposal to embrace more diversified, forward looking business models. This strategic shift acts as a signalling mechanism to the market, indicating a move towards a more sustainable and resilient business model.

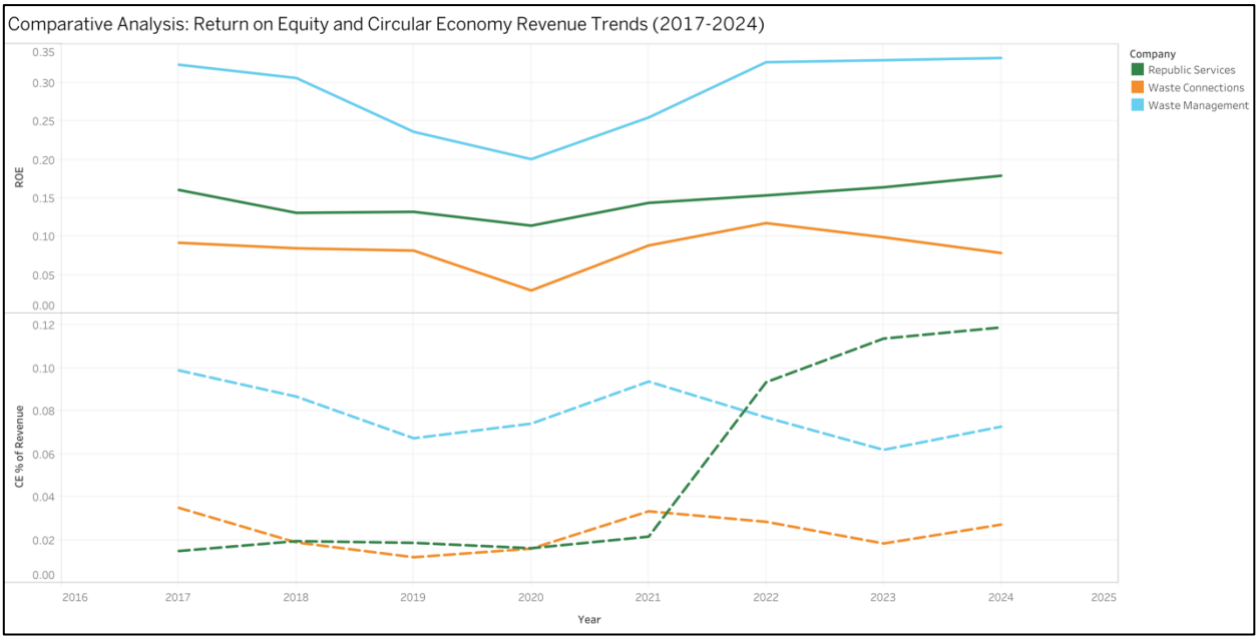


Figure 4: ROE and CE Trends

An analysis of Return on Equity (ROE) offers a deeper insight into how efficiently each company uses shareholder capital to generate profit. The data shows that Waste Management has the highest ROE, with a figure of 33.26% in 2024, which is consistently higher than both its competitors. This reflects WM's exceptional efficiency in leveraging its massive asset base to generate a return for shareholders. Republic Services' ROE of 17.92% in 2024 is strong but consistently lower than WM's. Meanwhile, Waste Connections has the lowest ROE, at 7.86% in 2024, which aligns with its lower net income percentage and its capital light, growth focused strategy (Appendix, Table 7.4.3). This lower ROE is a direct consequence of its strategic decision to reinvest earnings to fuel future growth, a choice the market rewards with a higher P/E multiple. This presents a key anomaly in the data: while WCN's lower ROE might seem counterintuitive to its premium valuation, it is a clear reflection of the market's willingness to value future growth over current profitability, a classic pattern in finance (Brealey, Myers and Allen, 2020).

Total Shareholder Return (TSR) provides the ultimate measure of shareholder value, capturing both capital gains and dividends. As shown by the comparative TSR from the Bloomberg data (Appendix, Figure 7.4.2), which presents a nuanced picture of long term investor rewards. While Waste Connections commanded a higher EV/EBITDA multiple and showcased a growth oriented strategy, its TSR performance was not consistently superior to its peers. Waste Management and Republic Services, despite their lower valuation multiples, have often delivered competitive or even higher TSR over the long run. For example, from 2017 to 2024, RSG delivered a TSR of 500%, WM delivered 380%, and WCN delivered 350%. This finding suggests that while the market may perceive WCN as a higher growth company and value it accordingly, the total returns for investors in WM and RSG, companies with a more traditional, asset heavy model are robust and reliable.

This higher TSR for WM and RSG directly challenges the simple narrative that CE focused, asset light models are the only path to long-term value, showing that the market continues to reward the stability and profitability of asset heavy businesses.

4.4 Conclusion of Analysis

The common-size financial analysis and company report comparison demonstrate two distinct, successful business models. Waste Management and Republic Services operate a traditional, capital intensive model built on landfill ownership, with a growing circular economy revenue stream and high ROE. In contrast, Waste Connections pursues a more agile, less capital intensive growth strategy, which the market rewards with a premium valuation. While different strategic paths can lead to profitability, the analysis of TSR reveals that robust returns are generated not only through growth oriented strategies but also through the steady profitability and stability of the asset heavy model, showing that both can lead to strong shareholder value.

Chapter 5: Discussion

This chapter analyses the study's findings, explaining them using the theoretical frameworks from the literature review. The main goal is to answer the research question: “Do circular economy strategies and landfill ownership contribute to long term shareholder value in global waste management firms?” The results show that both factors are important, but their contribution depends on the firm’s overall business model. This chapter argues there are two distinct, and equally successful, strategic approaches, each with its own way of generating shareholder value.

5.1 Landfill Ownership as a Differentiated Strategic Asset

The findings show a clear link between the ownership of landfill assets and long-term financial performance for firms like Waste Management (WM) and Republic Services (RSG). This relationship fits with the Resource-Based View (RBV) (Barney, 2007), which argues that a lasting competitive advantage comes from resources that are valuable, rare, hard to copy, and irreplaceable. Landfills are very hard to build because of many rules and logistical challenges. Analysis of the firms' asset portfolios highlights a clear and intentional accumulation of landfill assets. This strategy strengthens their market position and creates a strong economic "moat."

Owning these assets allows for a high degree of vertical integration, letting the companies gain value from the whole waste process, from collection to disposal. This control makes them less vulnerable to market changes and provides steady revenue, as shown in the companies' annual reports where collection and landfill operations dominate the revenue mix. This vertical integration also aligns with the principles of Transaction Cost Economics (TCE) (Williamson, 1985), which demonstrates how internalizing services reduces risks and lowers transaction costs. For investors, this asset heavy model signals operational stability and predictability, which is often rewarded with a robust and consistent Return on Equity (ROE) and Total Shareholder Return (TSR). This behaviour is supported by Materiality Theory, which suggests that investors prioritize issues that have a direct and tangible impact on a company's financial profile (Khan, Serafeim & Yoon, 2016). While this strategy is capital intensive, the secure, long-term returns it generates show that physical assets can be a powerful way to create and keep shareholder value. This reinforces the notion that asset concentration, when aligned with market perception, can amplify a firm's valuation.

5.2 Strategic Agility and Premium Market Valuation

In contrast to the asset heavy model, the evidence from Waste Connections (WCN) points to a different strategy for creating value. WCN’s business model, with its lower reliance on owned landfills, has been highly valued by the market, as shown by its high Enterprise Value to EBITDA (EV/EBITDA) multiple. This result supports Signaling Theory (Spence, 1973), which states that a company’s business decisions and financial structure show its real value and future potential to

investors. WCN's asset light approach signals strategic agility, a lower capital intensity, and a focus on growth through targeted acquisitions in fragmented, high margin markets.

The market's premium valuation of WCN appears to be a reward for a different kind of "moat," one based on a key strategic capability rather than a physical resource. This finding challenges a strict interpretation of the RBV, suggesting that a firm's competitive advantage can come from the ability to adapt and innovate, as explored by the Dynamic Capabilities theory (Teece, 1997). The company's consistent communication of its long-term strategic vision and strong ESG performance sends a powerful signal to investors, as academic research confirms that such non-financial information can act as a positive signal for future value and stability (Friede, Busch & Bassen, 2015). The market's willingness to pay a premium for WCN's perceived growth potential indicates that value is increasingly based on future opportunities, not just on the assets a company has now. This suggests that financial returns, not just a sustainability commitment, drive shareholder confidence.

The findings present a key insight: the Resource-Based View is a powerful framework for explaining value creation from tangible, scarce assets like landfills, while Signaling Theory is more effective at explaining how the market perceives and rewards intangible strategies, such as WCN's focus on growth and agility. In essence, WM and RSG derive their value from what they own, while WCN's value is driven by what the market perceives it is capable of achieving.

5.3 The Confounding Role of Circular Economy Strategies

This study finds that while all three firms have integrated Circular Economy (CE) strategies, these efforts don't have a simple, direct effect on shareholder value (Geissdoerfer et al., 2017). All three firms have adopted CE initiatives to varying degrees, with revenue from recycling and other recovery operations represented in the firms' annual and ESG reports. The findings suggest that the market doesn't give a higher value to a company just because it is committed to these initiatives, which addresses a common criticism in academic literature regarding the gap between CE theory and practical application (Pomponi & Moncaster, 2016; Korhonen et al., 2018).

Instead, CE strategies appear to be evaluated within the context of the firm's dominant business model. For Waste Management, CE initiatives are a natural extension of its existing, asset heavy infrastructure, allowing it to create new value streams and improve its resource efficiency. This is consistent with Triguero, Moreno-Mondéjar and Sáez-Martínez, 2023 who noted that sustainability can be a source of competitive advantage when it is integrated into a firm's strategy. In the case of Waste Connections, CE revenues are a part of a larger, growth oriented portfolio. The market's higher valuation of WCN is not driven primarily by its CE efforts, but by the perceived strategic agility and growth potential of its overall business model. This suggests that while CE is a necessary part of modern waste management strategy, its contribution to shareholder value mostly

depends on how it is used to strengthen a company's main competitive advantage, be it through asset control or strategic agility. It is a necessary part of the strategy, but not enough on its own to get a premium valuation. This highlights a need for a more sophisticated valuation model for non-financial assets.

5.4 Summary of Discussion

This chapter concludes that two distinct strategic pathways lead to long-term shareholder value in the waste management industry. The first is an asset heavy model, exemplified by Waste Management and Republic Services, where value is primarily derived from the control of rare physical assets, a finding consistent with the Resource-Based View. The second is an asset light, agility focused model, as demonstrated by Waste Connections, where value is created through dynamic capabilities and market signaled growth potential. The study concludes that circular economy initiatives add shareholder value primarily by strengthening a firm's existing strategic model, rather than acting as a standalone source of value. This finding emphasizes that a coherent and integrated corporate strategy is essential for shaping positive financial outcomes.

Chapter 6: Conclusion

6.1 Summary of Findings

This dissertation provides empirical evidence that tangible strategic assets, such as landfills, are a more consistent and measurable driver of shareholder value than intangible circular economy (CE) strategies. It finds that CE strategies often play an important supporting role in a firm's growth narrative. The dissertation has demonstrated that landfill ownership and circular economy strategies contribute to shareholder value through two distinct and equally viable strategic pathways. The comparative analysis of Waste Management (WM), Republic Services (RSG), and Waste Connections (WCN) revealed two successful business models. The first is an asset heavy, capital intensive model, demonstrated by WM and RSG, which relies on extensive landfill infrastructure. The second is a more agile, asset light model, as demonstrated by WCN, which focuses on strategic acquisitions and operational efficiency.

The empirical analysis provided clear quantitative evidence for these strategic divergences, with WM and RSG showing high depreciation, depletion, and amortization (DD&A) expenses, while WCN consistently commanded a premium market valuation. The study also found a nuanced role for circular economy strategies. For WM and RSG, CE activities constituted a substantial portion of revenue but did not, on their own, guarantee a higher valuation. For Waste Connections, CE revenue was a smaller portion of its total, yet the company still received a premium. This suggests that the market values CE strategies not in isolation but as a component of a company's overall business model. The final analysis of Total Shareholder Return (TSR) provided a critical insight: despite their different strategies, all three firms delivered robust long-term returns to shareholders, highlighting that different paths can lead to comparable ultimate financial success.

6.2 Answering the Research Question and Contributions

The central research question was: “Do circular economy strategies and landfill ownership contribute to long-term shareholder value in global waste management firms?” Based on the empirical findings, the answer is a qualified yes, but with a critical caveat. The contribution of these factors is not uniform; rather, it is entirely dependent on the firm's overarching strategy.

Landfill ownership directly contributes to shareholder value by creating a durable competitive "moat," consistent with the Resource-Based View (RBV) of the firm (Barney, 2007). For asset heavy firms, these assets create predictable cash flows and operational stability, which are rewarded by investors with stable valuations. Similarly, circular economy strategies contribute to shareholder value, but primarily as a complement to the core business model. For asset heavy firms, CE initiatives reinforce their existing strengths, creating new revenue streams and enhancing efficiency. In the case of Waste Connections, CE strategies contribute to value by supporting a

high growth narrative that is highly valued by the market, as predicted by Signaling Theory (Spence, 1973).

This study makes several key contributions. Academically, it provides empirical, firm level evidence that links tangible asset ownership and intangible circular strategies to shareholder value, enriching the application of the Resource-Based View and Signaling Theory. From a practical perspective, the findings have direct implications for investment decisions, highlighting that a singular focus on a firm's environmental scores may lead to a flawed valuation.

6.3 Final Thoughts, Limitations, and Future Research

The waste management industry is at an inflection point, with the linear "take-make-dispose" model giving way to a more circular one. This study confirms that this transition is well underway and is a key determinant of long-term value. However, the core strategic importance of landfills will not diminish. Their scarcity and the high barriers to entry ensure that they will remain central to the business models of the largest players. The true value lies not in a single asset or strategy, but in the intelligent and coherent integration of all a firm's choices.

This study, while valuable, has two primary limitations. The first is the small sample size, as the analysis was confined to three publicly traded firms. The second limitation is the reliance on publicly reported financial data, which often lacks the granularity needed to fully disaggregate CE specific revenues and costs. Future research could expand the sample size and use a more detailed dataset to provide a more precise analysis.

6.4 Broader Implications: Beyond Shareholder Value

While this study was shareholder-centric, the findings have significant broader implications for a wider range of stakeholders. For instance, circular economy strategies, even when they do not immediately translate into a premium stock valuation, can offer substantial long-term benefits that are not fully captured by financial metrics. The adoption of CE initiatives can reduce a firm’s environmental liability and reputational risk, while also building a stronger brand identity with a socially conscious consumer base (VisiumKMS, 2024). These long-term benefits contribute to societal well-being and are crucial for the industry's social license to operate (Ivanov, 2025). A company’s commitment to sustainability, beyond its effect on share price, can foster trust with local communities, enhance employee recruitment and retention, and open new avenues for collaboration with regulators and non-governmental organizations (Rendtorff, 2018). A holistic view of value creation in this industry must therefore extend beyond the stock market to include its impact on customers, employees, communities, and the environment.

Chapter 7: Appendix

7.1 Waste Management Inc.

Table 7.1.1: Income Statement

Income Statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023
Operating revenues	14,485.00	14,914.00	15,455.00	15,218.00	17,931.00	19,698.00	20,426.00
Income from operations (EBIT)	2,636.00	2,789.00	2,706.00	2,434.00	2,965.00	3,365.00	3,575.00
Consolidated net income	1,949.00	1,923.00	1,671.00	1,496.00	1,817.00	2,240.00	2,276.00
Effective tax rates	0.11	0.19	0.21	0.21	0.23	0.23	0.25
Depreciation, depletion and amortization	1,376.00	1,477.00	1,574.00	1,671.00	1,999.00	2,038.00	2,071.00
EBITDA	4,012.00	4,266.00	4,280.00	4,105.00	4,964.00	5,403.00	5,646.00
Diluted Earnings Per Share (DPS)	4.41	4.45	3.91	3.52	4.29	5.39	5.66

Table 7.1.2: Common-Size Income Statement Analysis

	Common Size							
Income Statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Operating revenues	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Income from operations (EBIT)	18.20%	18.70%	17.51%	15.99%	16.54%	17.08%	17.50%	18.42%
Consolidated net income	13.46%	12.89%	10.81%	9.83%	10.13%	11.37%	11.14%	12.44%
Effective tax rates	-	-	-	-	-	-	-	-
Depreciation, depletion and amortization	9.50%	9.90%	10.18%	10.98%	11.15%	10.35%	10.14%	10.28%
EBITDA	27.70%	28.60%	27.69%	26.97%	27.68%	27.43%	27.64%	28.69%
Diluted Earnings Per Share (DPS)	-	-	-	-	-	-	-	-

Table 7.1.3: Balance Sheet

Balance Sheet (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total assets	21,829.00	22,650.00	27,743.00	29,345.00	29,097.00	31,367.00	32,823.00	44,567.00
TSH - Total Waste Management, Inc. stockholders	6,019.00	6,275.00	7,068.00	7,452.00	7,124.00	6,849.00	6,903.00	8,252.00
Landfill and environmental remediation liabilities	1,770.00	1,828.00	1,930.00	2,222.00	2,373.00	2,700.00	2,888.00	3,048.00
Current Liabilities	1,040.00	1,037.00	3,144.00	3,553.00	4,082.00	1,766.00	4,226.00	6,258.00
Total Debt	9,491.00	10,026.00	13,498.00	13,810.00	13,405.00	14,984.00	16,229.00	23,900.00
Final capping, closure, post-closure and environmental remediation funds	101.00	103.00	109.00	114.00	118.00	113.00	119.00	128.00
PP&E: Landfills	14,904	15,240	15,910	16,842	17,734	18,526	19,473	20,273

Table 7.1.4: Common-Size Balance Sheet Analysis

	Common Size							
Balance Sheet (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total assets	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
TSH - Total Waste Management, Inc. stockholders	27.57%	27.70%	25.48%	25.39%	24.48%	21.84%	21.03%	18.52%
Landfill and environmental remediation liabilities	8.11%	8.07%	6.96%	7.57%	8.16%	8.61%	8.80%	6.84%
Current Liabilities	4.76%	4.58%	11.33%	12.11%	14.03%	5.63%	12.88%	14.04%
Total Debt	43.48%	44.26%	48.65%	47.06%	46.07%	47.77%	49.44%	53.63%
Final capping, closure, post-closure and environmental remediation funds	0.46%	0.45%	0.39%	0.39%	0.41%	0.36%	0.36%	0.29%
PP&E: Landfills	68.28%	67.28%	57.35%	57.39%	60.95%	59.06%	59.33%	45.49%

Table 7.1.5: Cashflow Statement

Cashflow statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Net cash provided by operating activities	3,180.00	3,570.00	3,874.00	3,403.00	4,338.00	4,536.00	4,719.00	5,390.00
Capital expenditures	-1,509.00	-1,694.00	-1,818.00	-1,632.00	-1,904.00	-2,587.00	-2,895.00	-3,231.00
FCF	4,689.00	5,264.00	5,692.00	5,035.00	6,242.00	7,123.00	7,614.00	8,621.00
Total Dividends (Cash dividends)	750.00	802.00	876.00	927.00	970.00	1,077.00	1,136.00	1,210.00

Table 7.1.6: Total Shareholder Return and Share Price Analysis

Equity	2016	2017	2018	2019	2020	2021	2022	2023	2024
Total Shareholder return calculated	-	10.79	9.32	10.12	8.17	8.64	6.39	7.38	6.88
Share price on 31st December (closing price)	70.91	86.3	88.99	113.96	117.93	166.9	156.88	179.1	201.79

Table 7.1.7: Comparative Return on Stock and Index

Return on Index/Stock		Indexed Returns for the Years Ended December 31,				
Five year total return cumulative	2019 (Base Yr.)	2020	2021	2022	2023	2024
Waste Management, Inc	100	105	152	145	168	192
S&P 500 Index	100	118	152	125	158	197
Dow Jones Waste & Disposal Services Index	100	107	149	141	166	198

Table 7.1.8: Operating Revenue Segments

Net Operating Revenues (Dollars in millions, Year Ended December 31)	2017	2018	2019	2020	2021	2022	2023	2024
Total collection (Commercial, Industrial, Residential, Other)	9,264.00	9,724.00	10,240.00	10,053.00	11,675.00	13,615.00	14,356.00	14,890.00
Landfill	3,370.00	3,560.00	3,846.00	3,667.00	4,153.00	3,062.00	3,252.00	3,445.00
Transfer	1,591.00	1,711.00	1,820.00	1,855.00	2,072.00	1,166.00	1,257.00	1,381.00
Recycling Processing and Sales	1,432.00	1,293.00	1,040.00	1,127.00	1,681.00	1,516.00	1,264.00	1,603.00

Table 7.1.9: Common-Size Analysis of Operating Revenue by Segment

	Common Size							
Net Operating Revenues (Dollars in millions, Year Ended December 31)	2017	2018	2019	2020	2021	2022	2023	2024
Total collection (Commercial, Industrial, Residential, Other)	63.96%	65.20%	66.26%	66.06%	65.11%	69.12%	70.28%	67.49%
Landfill	23.27%	23.87%	24.89%	24.10%	23.16%	15.54%	15.92%	15.61%
Transfer	10.98%	11.47%	11.78%	12.19%	11.56%	5.92%	6.15%	6.26%
Recycling Processing and Sales	9.89%	8.67%	6.73%	7.41%	9.37%	7.70%	6.19%	7.27%

Table 7.1.10: Operating Cost Segments

Operating Cost	2017	2018	2019	2020	2021	2022	2023	2024
Total landfill operating costs	328.00	331.00	379.00	394.00	412.00	421.00	453.00	524.00

Table 7.1.11: Owned and Operated Landfills

Owned and operated landfills	2017	2018	2019	2020	2021	2022	2023	2024
0 to 5 years	26	30	27	26	28	28	31	28
6 to 10 years	21	16	16	23	21	23	22	22
11 to 20 years	27	35	39	45	50	53	50	53
21 to 40 years	74	68	65	63	61	62	66	62
41+ years	101	103	102	111	100	93	94	97
Total	249	252	249	268	260	259	263	262

Table 7.1.12: ESG and Circularity Metrics

ESG & Circularity Metrics	2017	2018	2019	2020	2021	2022	2023	2024
Landfill Gas Converted to Energy for Sale/Use (MMBTU)	56,960,000.00	58,630,000.00	58,060,000.00	56,130,000.00	55,510,000.00	54,504,000.00	56,621,600.00	58,096,500.00
Total Waste Generated (Metric Tons)	3,964.00	4,032.00	4,138.00	3,588.00	4,469.00	4,561.00	4,400.00	4,500.00
Total Waste Recycled (Metric Tons)	1,075.00	1,068.00	1,099.00	904.00	1,185.00	1,204.00	1,200.00	1,200.00
Circularity Tons recovered	15,330,000.00	15,161,672.00	15,510,697.00	15,030,654.00	15,337,456.00	14,831,559.00	15,237,738.00	16,037,600.00
Landfill Gas Captured (MTCO2e)	49,010,808.00	49,655,869.00	51,088,479.00	51,136,297.00	53,562,528.00	53,408,981.00	53,941,100.00	57,810,500.00
Landfill Gas Recovered for Beneficial Use	54%	52%	53%	51%	45%	45%	43%	45%

Table 7.1.13: Financial Ratios

Financial Ratios	2017	2018	2019	2020	2021	2022	2023	2024
Return on Equity	32.38%	30.65%	23.64%	20.08%	25.51%	32.71%	32.97%	33.26%
P/E Ratio	19.57	20.00	29.15	33.50	38.90	29.11	31.64	29.63

Table 7.1.14: Bloomberg Financial Data

Bloomberg Data	2017	2018	2019	2020	2021	2022	2023	2024
EV/EBITDA	3.24	3.20	3.80	4.18	4.64	4.02	4.32	4.79
EV/Sales	11.69	3.20	3.80	4.18	4.64	4.02	4.32	4.79
BESG Environmental Pillar Score	2.30	4.73	4.89	4.77	4.25	2.95	2.82	-
Environmental Disclosure Score	60.01	58.29	60.01	60.01	61.16	61.16	67.93	-
BESG ESG Score	4.32	5.66	5.64	5.77	5.63	4.83	4.08	-
GHG Scope 1	15690.00	16272.10	15624.60	16083.20	16975.30	15321.70	14938.70	13223.30
GHG Scope 2	244.83	246.09	238.34	236.15	257.19	301.88	285.37	
GHG Scope 3	11009.30	11205.40	3211.66	3502.99	3420.41	1488.09	1811.75	1953.50
Renewable Electricity/Energy Generated	282.84	571.34	1210.24	1993.60	1952.44	2029.68	2707.16	
FCF Yield	4.41	4.91	4.25	3.55	3.47	3.01	2.52	2.66
Shareholder Yield	3.64	4.00	-4.07	3.12	4.17	1.91	2.12	-6.35
PE Ratio	25.8092	21.29	25.8343	29.2567	34.461	28.0631	29.1489	28.4438

Table 7.1.15: LSEG Financial Data

LSEG Data	2017	2018	2019	2020	2021	2022	2023	2024
ESGScore	A	A	A	A	A	A	A	A
ESGCombinedScore	A	A	A	A	A	C+	A	A
EnvironmentPillarScore	A+	A+	B+	B+	B+	B+	B	B
GovernancePillarScore	A	A	A+	A+	A+	A	A+	A+
SocialPillarScore	A	A	A+	A+	A	A	A	A

Table 7.1.16: Landfill Assets

	2017			2018			2019		
Landfill Assets	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**
Previous Year Ended December 31	14,276.00	-8,340.00	5,936.00	14,904.00	-8,788.00	6,116.00	15,240.00	-9,157.00	6,083.00
Capital additions	498.00	-	498.00	513.00	-	513.00	656.00	-	656.00
Asset retirement obligations incurred and capitalized	100.00	-	100.00	83.00	-	83.00	72.00	-	289.00
Acquisitions	84.00	-	84.00	2.00	-	2.00	289.00	-	289.00
Depletion of landfill airspace	-	-497.00	-497.00	-	-538.00	-538.00	-	-575.00	-575.00
Foreign currency translation	71.00	-25.00	46.00	-89.00	35.00	-54.00	52.00	-22.00	30.00
Asset retirements and other adjustments	-125.00	74.00	-51.00	-173.00	134.00	-39.00	-399.00	428.00	29.00
Current Year Ended December 31	14,904.00	-8,788.00	6,116.00	15,240.00	-9,157.00	6,083.00	15,910.00	-9,326.00	6,584.00

	2020			2021			2022		
Landfill Assets	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**
Previous Year Ended December 31	15,910.00	-9,326.00	6,584.00	16,842.00	-9,692.00	7,150.00	17,734.00	-10,390.00	7,344.00
Capital additions	546.00	-	546.00	791.00	-	791.00	791.00	-	791.00
Asset retirement obligations incurred and capitalized	83.00	-	83.00	117.00	-	117.00	114.00	-	114.00
Acquisitions	607.00	-	607.00	-	-	-	-	-	-
Depletion of landfill airspace	-	-568.00	-568.00	-	-731.00	-731.00	-	-754.00	-754.00
Foreign currency translation	22.00	-10.00	12.00	8.00	-3.00	5.00	-81.00	36.00	-45.00
Asset retirements and other adjustments	-326.00	212.00	-114.00	-24.00	36.00	12.00	-32.00	212.00	180.00
Current Year Ended December 31	16,842.00	-9,692.00	7,150.00	17,734.00	-10,390.00	7,344.00	18,526.00	-10,896.00	7,630.00

	2023			2024		
Landfill Assets	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**	**Cost Basis of Landfill Assets**	**Accumulated Landfill Airspace Depletion**	**Net Book Value of Landfill Assets**
Previous Year Ended December 31	18,526.00	-10,896.00	7,630.00	19,473.00	-11,643.00	7,830.00
Capital additions	722.00	-	722.00	832.00	-	832.00
Asset retirement obligations incurred and capitalized	79.00	-	79.00	91.00	-	91.00
Acquisitions	-	-	-	-	-	-
Depletion of landfill airspace	-	-745.00	-745.00	-	-795.00	-795.00
Foreign currency translation	28.00	-12.00	16.00	-100.00	47.00	-53.00
Asset retirements and other adjustments	118.00	10.00	128.00	-23.00	132.00	109.00
Current Year Ended December 31	19,473.00	-11,643.00	7,830.00	20,273.00	-12,259.00	8,014.00

Table 7.1.17: Landfill Airspace Depletion Expense

Landfill Airspace Depletion Expense	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Depletion of landfill airspace (in millions)	497.00	538.00	575.00	568.00	731.00	754.00	745.00	795.00
Tons received, net of redirected waste (in millions)	112.00	116.00	121.00	112.00	124.00	125.00	123.00	125.00
Average landfill airspace depletion expense per ton	4.44	4.64	4.75	5.07	5.9	6.05	6.07	6.36

7.2 Waste Connections

Table 7.2.1: Income Statement

Income Statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Revenue	4,630.49	4,922.94	5,388.68	5,445.99	6,151.36	7,211.86	8,021.95	8,919.59
Operating Income (EBIT)	627.10	832.24	837.78	412.43	1,039.63	1,242.19	1,236.31	1,067.66
Net income attributable to WCN	576.82	546.87	566.84	204.68	618.05	835.66	762.80	617.57
Effective tax rate	-13.60%	22.60%	19.70%	19.70%	19.80%	20.30%	22.40%	19.20%
Depreciation, depletion and amortization	632.48	680.49	743.92	752.40	813.01	918.96	1,003.21	1,163.77
EBITDA	2,093.01	2,246.91	2,417.47	2,414.39	2,732.17	3,139.61	3,526.17	4,065.63
Diluted Earnings Per Share (DPS)	2.18	2.07	2.14	0.78	2.36	3.24	2.95	2.39

Table 7.2.2: Common-Size Income Statement Analysis

	Common Size							
Income Statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Revenue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Operating Income (EBIT)	13.54%	16.91%	15.55%	7.57%	16.90%	17.22%	15.41%	11.97%
Net income attributable to WCN	12.46%	11.11%	10.52%	3.76%	10.05%	11.59%	9.51%	6.92%
Effective tax rate	-	-	-	-	-	-	-	-
Depreciation, depletion and amortization	13.66%	13.82%	13.81%	13.82%	13.22%	12.74%	12.51%	13.05%
EBITDA	45.20%	45.64%	44.86%	44.33%	44.42%	43.53%	43.96%	45.58%
Diluted Earnings Per Share (DPS)	-	-	-	-	-	-	-	-

Table 7.2.3: Balance Sheet

Balance Sheet (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total Assets	12,014.68	12,627.33	13,737.70	13,992.36	14,699.92	17,134.60	17,915.88	19,817.81
Total Equity	6,274.07	6,460.19	6,938.35	6,863.44	6,993.55	7,113.64	7,697.78	7,860.35
Landfill closure/post-closure: Deferred income tax Assets/Liabilities	-7.76	-8.89	-12.16	-13.85	-18.60	-22.75	12.48	90.24
Current Liabilities	802.60	860.71	1,007.23	1,028.65	1,232.75	1,512.64	1,687.93	1,878.83
Total Debt	3,911.23	4,155.25	4,354.25	4,716.95	5,046.52	6,896.91	6,751.23	8,080.78

Table 7.2.4: Common-Size Balance Sheet Analysis

	Common Size							
Balance Sheet (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total Assets	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Total Equity	52.22%	51.16%	50.51%	49.05%	47.58%	41.52%	42.97%	39.66%
Landfill closure/post-closure: Deferred income tax Assets/Liabilities	-0.06%	-0.07%	-0.09%	-0.10%	-0.13%	-0.13%	0.07%	0.46%
Current Liabilities	6.68%	6.82%	7.33%	7.35%	8.39%	8.83%	9.42%	9.48%
Total Debt	32.55%	32.91%	31.70%	33.71%	34.33%	40.25%	37.68%	40.78%

Table 7.2.5: Cashflow Statement

Cashflow statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Net cash provided by operating activities	1,187.26	1,411.24	1,540.55	1,408.52	1,698.23	2,022.49	2,126.82	2,228.93
Capital expenditures for property and equipment	479.29	546.15	-634.41	597.05	744.32	912.68	934.00	1,055.99
Adjusted Free Cash Flow	763.89	879.90	916.78	841.90	1,009.60	1,164.84	1,224.13	1,217.53
Capping, closure and post-closure expenditures	-8.85	-2.70	-5.06	-6.48	-21.04	-18.88	-39.43	-247.94
Total Dividends (Cash dividends)	131.98	152.55	175.07	199.88	220.20	243.01	270.60	302.26

Table 7.2.6: Total Shareholder Return and Share Price Analysis

Return on Index/Stock		Indexed Returns for the Years Ended December 31,				
Company Name / Index - dollars	2019 (Base Yr.)	2020	2021	2022	2023	2024
Waste Connections, Inc	100	$113.86	$152.33	$149.25	$169.36	$195.99
S&P 500 Index	100	$118.40	$152.39	$124.79	$157.59	$197.02
Dow Jones U.S. Waste & Disposal Services Index	100	$106.56	$148.97	$140.91	$165.99	$197.91

Table 7.2.7: Comparative Return on Stock and Index

Equity	2016	2017	2018	2019	2020	2021	2022	2023	2024
Total Shareholder return calculated	-	2.87%	2.20%	2.58%	2.33%	2.48%	1.76%	2.17%	2.17%
Share price (closing price)	$52.39	$70.94	$74.25	$90.79	$102.57	$136.27	$132.56	$149.27	$171.58

Table 7.2.8: Operating Revenue Segments

Revenue Segment (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total collection (Commercial, Residential, Industrial and construction roll off)	3,181.45	3,410.67	3,815.15	3,971.68	4,441.43	5,251.03	5,934.98	6,332.77
Landfill	988.09	1,063.24	1,132.94	1,146.73	1,233.50	1,328.94	1,483.40	1,557.87
Transfer	589.88	670.13	771.32	777.75	859.11	1,026.05	1,198.39	1,349.08
Recycling	161.73	92.63	64.25	86.39	205.08	204.88	147.04	241.87
E&P	203.47	256.26	271.89	159.44	138.71	210.56	232.21	521.50
Intermodal and other	146.75	139.90	121.14	118.40	152.19	188.47	171.72	191.89
Intercompany	-640.89	-709.89	-787.99	-814.40	-878.65	-998.07	-1,145.78	-1,275.40
Total	4,630.49	4,922.94	5,388.68	5,445.99	6,151.36	7,211.86	8,021.95	8,919.59

Table 7.2.9: Common-Size Analysis of Operating Revenue by Segment

	Common Size							
Revenue Segment (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total collection (Commercial, Residential, Industrial and construction roll off)	68.71%	69.28%	70.80%	72.93%	72.20%	72.81%	73.98%	71.00%
Landfill	21.34%	21.60%	21.02%	21.06%	20.05%	18.43%	18.49%	17.47%
Transfer	12.74%	13.61%	14.31%	14.28%	13.97%	14.23%	14.94%	15.12%
Recycling	3.49%	1.88%	1.19%	1.59%	3.33%	2.84%	1.83%	2.71%
E&P	4.39%	5.21%	5.05%	2.93%	2.25%	2.92%	2.89%	5.85%
Intermodal and other	3.17%	2.84%	2.25%	2.17%	2.47%	2.61%	2.14%	2.15%
Intercompany	-13.84%	-14.42%	-14.62%	-14.95%	-14.28%	-13.84%	-14.28%	-14.30%
Total	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Table 7.2.10: Owned and Operated Landfills

Owned and operated landfills	2017	2018	2019	2020	2021	2022	2023	2024
0 to 5	6	4	7	5	3	5	7	6
6 to 10	3	7	6	2	6	5	3	7
11 to 20	15	19	18	18	15	18	21	25
21 to 40	34	30	32	36	37	39	36	37
41 to 50	9	10	8	5	9	6	9	11
51+	19	19	21	21	22	20	20	20
Total	86	89	92	87	92	93	96	106

Table 7.2.11: ESG and Circularity Metrics

ESG & Circularity Metrics	2017	2018	2019	2020	2021	2022	2023	2024
Landfill site costs	3,606.43	3,981.87	4,334.56	4,205.97	4,447.53	5,179.77	5,507.60	5,778.48
Final capping, closure and post-closure liability at December 31	237.82	251.78	291.47	301.90	302.54	344.61	522.23	860.12
Amount of material recycled (Metric tons (t))	Report Not Available	1,394.40	1,541.79	1,615.54	1,673.44	1,981.18	2,002.63	Not released data
Total landfill gas recovered (Million British thermal units (MMBtu))	Report Not Available	18,476.40	18,521.77	22,939.94	23,600.22	23,621.52	25,267.52	Not released data
Landfill gas used for energy	Report Not Available	41%	61%	60%	61%	60%	58%	Not released data
Landfill Tons (tons per year)	41,159,167.00	45,520.00	47,964.00	44,890.00	47,212.00	47,849.00	50,407.00	Not released data

Table 7.2.12: Financial Ratios

Financial Ratios	2017	2018	2019	2020	2021	2022	2023	2024
Return on Equity	9.19%	8.47%	8.17%	2.98%	8.84%	11.75%	9.91%	7.86%
P/E Ratio	$32.54	$35.87	$42.43	$131.50	$57.74	$40.91	$50.60	$71.79

Table 7.2.13: Bloomberg Financial Data

Bloomberg Data	2017	2018	2019	2020	2021	2022	2023	2024
EV/EBITDA	17.84	15.3	17.53	26.1	21.59	18.82	20.05	23.15
EV/Sales	4.85	4.8	5.27	5.77	6.64	5.75	5.72	5.93
BESG Environmental Pillar Score	0.29	2.29	3.19	3.19	3.36	4.18	4.19	-
Environmental Disclosure Score	5.5	7.22	12.96	16.16	34.79	34.79	49.17	-
BESG ESG Score	2.41	4.29	5.22	4.83	5.24	5.8	5.95	-
GHG Scope 1	9252.38	5089.08	6006.64	5609.96	5600.96	4812.96	5237.96	-
GHG Scope 2	103.15	53.99	55.44	51.51	50.69	49.81	56.12	-
GHG Scope 3	1579.43	1710.81	1943.94	931.52	1052.59	1080.55	1015.84	-
FCF Yield	3.77	4.41	3.78	3.02	2.68	3.25	3.09	2.65
Shareholder Yield	-0.31	0.97	0.4	0.27	1.41	-3.01	1.41	-2.14
PE Ratio	38.6525	33.9767	38.6711	45.5706	48.3363	39.2749	39.876	39.506

Table 7.2.14: LSEG Financial Data

LSEG Data	2017	2018	2019	2020	2021	2022	2023	2024
ESGScore	C+	C+	C+	C+	C+	B-	B	B
ESGCombinedScore	C+	C+	C+	C+	C+	B-	C+	C+
EnvironmentPillarScore	C-	C+	C+	B-	B-	B-	B-	B-
GovernancePillarScore	A-	A-	B	B+	A-	A-	A	A+
SocialPillarScore	C-	C-	C-	C-	C-	C	C+	C

7.3 Republic Services

Table 7.3.1: Income Statement

Income Statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Revenue	10,041.50	10,040.90	10,299.40	10,153.60	11,295.00	13,511.30	14,964.50	16,032.00
Operating income (EBIT)	1,668.50	1,735.80	1,787.20	1,709.10	2,076.20	2,391.70	2,780.20	3,196.00
Net income	1,279.00	1,037.60	1,073.80	969.60	1,292.30	1,487.60	1,731.40	2,044.00
Effective income tax rate	0.20%	21.50%	17.10%	15.20%	18.00%	18.80%	21.00%	16.00%
Depreciation, depletion and amortization	1,036.30	1,033.40	1,040.50	1,075.90	1,185.50	1,352.00	1,501.00	1,677.00
EBITDA	2,704.80	2,769.20	2,827.70	2,785.00	3,261.70	3,743.70	4,281.20	4,873.00
Diluted Earnings Per Share (DPS)	3.77	3.16	3.33	3.02	4.04	4.69	5.47	6.49

Table 7.3.2: Common-Size Income Statement Analysis

	Common Size							
Income Statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Revenue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Operating income (EBIT)	16.62%	17.29%	17.35%	16.83%	18.38%	17.70%	18.58%	19.94%
Net income	12.74%	10.33%	10.43%	9.55%	11.44%	11.01%	11.57%	12.75%
Effective income tax rate	-	-	-	-	-	-	-	-
Depreciation, depletion and amortization	10.32%	10.29%	10.10%	10.60%	10.50%	10.01%	10.03%	10.46%
EBITDA	26.94%	27.58%	27.45%	27.43%	28.88%	27.71%	28.61%	30.40%
Diluted Earnings Per Share (DPS)	-	-	-	-	-	-	-	-

Table 7.3.3: Balance Sheet

Balance Sheet (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total assets	21,147.00	21,617.00	22,683.80	23,434.00	24,955.00	29,052.90	31,410.10	32,402.00
Total stockholders' equity	7,961.10	7,929.50	8,120.90	8,488.80	8,979.70	9,686.80	10,543.50	11,407.00
Accrued landfill and environmental costs, current portion	135.20	130.60	132.60	114.50	124.50	132.60	141.00	159.00
Accrued landfill and environmental costs, net of current portion	1686.50	1028.30	1180.60	1694.70	1837.70	2141.30	2281.00	2432.00
Current Liabilities	598.10	761.50	3064.60	2282.00	2415.60	3390.70	4228.00	4128.00
Total Debt	8,187	8,338	8,689	8,934	9,554	11,786	12,819	12,713
Landfill development costs	6,757.30	7,106.00	7,474.70	7,991.70	8,539.60	9,574.20	9,911.00	10,518.00
Construction-in-progress – landfill	233.20	287.90	366.80	303.80	279.30	358.30	350.00	437.00

Table 7.3.4: Common-Size Balance Sheet Analysis

	Common Size							
Balance Sheet (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total assets	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%
Total stockholders' equity	37.65%	36.68%	35.80%	36.22%	35.98%	33.34%	33.57%	35.20%
Accrued landfill and environmental costs, current portion	0.64%	0.60%	0.58%	0.49%	0.50%	0.46%	0.45%	0.49%
Accrued landfill and environmental costs, net of current portion	7.98%	4.76%	5.20%	7.23%	7.36%	7.37%	7.26%	7.51%
Current Liabilities	2.83%	3.52%	13.51%	9.74%	9.68%	11.67%	13.46%	12.74%
Total Debt	38.72%	38.57%	38.30%	38.12%	38.29%	40.57%	40.81%	39.24%
Landfill development costs	31.95%	32.87%	32.95%	34.10%	34.22%	32.95%	31.55%	32.46%
Construction-in-progress – landfill	1.10%	1.33%	1.62%	1.30%	1.12%	1.23%	1.11%	1.35%

Table 7.3.5: Cashflow Statement

Cashflow statement (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Cash provided by operating activities	1,910.70	2,242.80	2,352.10	2,471.60	2,786.70	3,190.00	3,617.80	3,936.00
Capital expenditures	989.80	1,071.80	1,207.10	1,194.60	1,316.30	1,454.00	1,631.10	1,855.00
Free Cash Flow	927.00	1,171.00	1,200.00	1,240.00	1,520.00	1,736.00	1,986.70	2,081.00
Total Dividends (Cash dividends)	461.80	491.20	491.20	522.50	552.60	593.00	638.00	687.00

Table 7.3.6: Total Shareholder Return and Share Price Analysis

		Indexed Returns for the Years Ended December 31,				
Return on Index/Stock	2019 (Base Yr.)	2020	2021	2022	2023	2024
Republic Services, Inc	100	$110	$161	$151	$196	$241
S&P 500 Index	100	$118	$152	$125	$158	$197
DJ W&DS Index	100	$107	$149	$141	$166	$198

Table 7.3.7: Comparative Return on Stock and Index

Equity	2016	2017	2018	2019	2020	2021	2022	2023	2024
Total Shareholder return calculated	-	8.28	7.33	7.06	5.90	6.19	4.18	5.22	4.39
Share price (closing price)	$57.05	$67.61	$72.09	$89.63	$96.30	$139.45	$128.99	$164.91	$201.18

Table 7.3.8: Operating Revenue Segments

Net Operating Revenues (Dollars in millions)	2017	2018	2019	2020	2021	2022	2023	2024
Total collection	7,413.40	7,524.10	7,737.60	7,589.50	8,278.20	9,343.30	10,252.90	10,855.00
Transfer	1,209.50	1,244.90	1,318.60	1,349.40	1,490.00	1,574.50	1,699.10	1,780.00
Less: intercompany	-703.80	-713.80	-748.00	-745.90	-814.40	-849.80	-933.70	-975.00
Transfer, net	505.70	531.10	570.60	603.50	675.60	724.70	765.40	805.00
Landfill	2,224.30	2,291.70	2,328.80	2,307.30	2,516.60	2,681.70	2,885.40	2,981.00
Less: intercompany	-985.50	-1,020.80	-1,028.70	-1,018.50	-1,092.80	-1,131.90	-1,206.00	-1,240.00
Landfill, net	1,238.80	1,270.90	1,300.10	1,288.80	1,423.80	1,549.80	1,679.40	1,741.00
Environmental solutions	149.00	194.70	191.80	163.30	242.40	1,262.10	1,701.40	1,907.00
Less: intercompany	-	-	-	-18.10	-19.50	-53.90	-76.50	-64.00
Environmental solutions, net	-	-	-	145.20	222.90	1,208.20	1,624.90	1,843.00
Other:								
Recycling processing and commodity sales	539.20	297.80	273.20	297.10	420.50	359.10	312.30	409.00
Other non-core	195.40	222.30	226.10	229.50	274.00	326.20	329.60	379.00
Total other	734.60	520.10	499.30	526.60	694.50	685.30	641.90	788.00
Total revenue	10,041.50	10,040.90	10,299.40	10,153.60	11,295.00	13,511.30	14,964.50	16,032.00

Table 7.3.9: Common-Size Analysis of Operating Revenue by Segment

	Common Size							
Net Operating Revenues (Dollars in millions)	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Total collection	73.83%	74.93%	75.13%	74.75%	73.29%	69.15%	68.51%	67.71%
Transfer	12.05%	12.40%	12.80%	13.29%	13.19%	11.65%	11.35%	11.10%
Less: intercompany	-7.01%	-7.11%	-7.26%	-7.35%	-7.21%	-6.29%	-6.24%	-6.08%
Transfer, net	5.04%	5.29%	5.54%	5.94%	5.98%	5.36%	5.11%	5.02%
Landfill	22.15%	22.82%	22.61%	22.72%	22.28%	19.85%	19.28%	18.59%
Less: intercompany	-9.81%	-10.17%	-9.99%	-10.03%	-9.68%	-8.38%	-8.06%	-7.73%
Landfill, net	12.34%	12.66%	12.62%	12.69%	12.61%	11.47%	11.22%	10.86%
Environmental solutions	1.48%	1.94%	1.86%	1.61%	2.15%	9.34%	11.37%	11.89%
Less: intercompany	-	-	-	-0.18%	-0.17%	-0.40%	-0.51%	-0.40%
Environmental solutions, net	-	-	-	1.43%	1.97%	8.94%	10.86%	11.50%
Other:	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Recycling processing and commodity sales	5.37%	2.97%	2.65%	2.93%	3.72%	2.66%	2.09%	2.55%
Other non-core	1.95%	2.21%	2.20%	2.26%	2.43%	2.41%	2.20%	2.36%
Total other	7.32%	5.18%	4.85%	5.19%	6.15%	5.07%	4.29%	4.92%
Total revenue	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%

Table 7.3.10: Operating Cost Segments

Operating Expenses	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Labor and related benefits	2,027.20	2,144.30	2,202.40	2,153.40	2,324.40	2,702.90	2,993.90	3,213.00
Transfer and disposal costs	795.90	829.30	841.70	796.90	865.80	992.90	1,056.30	1,101.00
Maintenance and repairs	940.20	986.60	1,006.20	969.60	1,048.80	1,228.40	1,388.30	1,468.00
Transportation and subcontract costs	585.80	647.60	674.90	674.10	779.50	1,086.50	1,171.00	1,212.00
Fuel	349.80	391.40	347.90	271.70	383.00	631.10	541.60	470.00
Disposal fees and taxes	468.90	322.00	325.70	313.50	336.60	342.30	347.90	351.00
Landfill operating costs	220.30	241.60	244.70	258.20	258.90	283.20	333.00	367.00
Risk management	212.60	217.90	230.70	213.90	261.60	321.40	385.20	401.00
Other	377.00	409.30	440.60	460.00	479.10	616.00	725.00	796.00
Subtotal	6,214.60	6,190.00	6,314.80	6,111.30	6,737.70	8,204.70	8,942.20	9,379.00
Total cost of operations	6,214.60	6,150.00	6,298.40	6,100.50	6,737.70	8,205.00	8,942.20	9,350.00

Table 7.3.11: Owned and Operated Landfills

Owned and operated landfills	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
0 to 5 years	16	12	11	15	21	23	21	23
6 to 10 years	18	18	20	16	19	20	22	22
11 to 20 years	35	31	32	29	28	31	36	38
21 to 40 years	49	55	53	52	58	55	54	54
41+ years	77	74	73	74	72	77	74	71
Total	195	190	189	186	198	206	207	208

Table 7.3.12: ESG and Circularity Metrics

ESG & Circularity Metrics	**2017**	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**	**2024**
Total landfill gas recovered MMBTU	78,942,711.00	80,148,584.00	82,275,736.00	83,701,380.00	83,135,235.00	78,411,864.00	77,620,558.00	77,439,287.00
Amount of material recycled (Million metric tons)	5.40	5.40	4.90	5.10	4.80	3.10	2.90	3.10
Amount of materials processed as waste to energy (Million metric tons)	43.00	45.00	44.50	43.00	40.80	42.20	44.40	46.80

Table 7.3.13: Financial Ratios

Financial Ratios	2017	2018	2019	2020	2021	2022	2023	2024
Return on Equity	16.07%	13.09%	13.22%	11.42%	14.39%	15.36%	16.42%	17.92%
P/E Ratio	17.93	22.81	26.92	31.89	34.52	27.50	30.15	31.00

Table 7.3.14: Bloomberg Financial Data

Bloomberg Data	2017	2018	2019	2020	2021	2022	2023	2024
EV/EBITDA	10.96	11.06	12.7	13.68	15.94	13.56	14.6	15.04
EV/Sales	3.04	3.14	3.64	3.92	4.78	3.9	4.33	4.72
BESG Environmental Pillar Score	2.28	2.44	3.12	4.39	4.7	4.55	5.04	-
Environmental Disclosure Score	31.32	31.32	31.32	48.23	59.44	59.44	60.28	-
BESG ESG Score	4.44	4.6	5.04	5.75	5.96	5.95	6.34	-
GHG Emission Scope 1	17022.8	15249.9	15037.8	14280.6	15499.2	15462	14622.5	13608.5
GHG Emission Scope 2	390.83	188.29	186.57	188.38	197.11	182.78	186.58	185.21
GHG Emission Scope 3	1841.53	2791.73	2809.24	2751.68	1923.91	1951.39	2010.69	2011.5
Renewable Electricity/Energy Generated	432.92	865.85	865.85	1146.18	1243.84	1222.25	1205.1	1423.77
FCF Yield	4.04	4.97	3.98	4.15	3.31	4.25	3.81	3.29
Shareholder Yield	2.29	4.56	2.06	1.99	0.74	-3.29	-0.12	2.23
PE Ratio	27.8373	23.3336	27.1452	26.9969	34.012	26.1911	29.5144	31.1208

Table 7.3.15: LSEG Financial Data

LSEG Data	2017	2018	2019	2020	2021	2022	2023	2024
ESGScore	B+	A-	B+	B+	B+	A-	A-	A-
ESGCombinedScore	B+	A-	B	B+	B+	A-	A-	A-
EnvironmentPillarScore	A-	A	A	A-	A	A+	A	A
GovernancePillarScore	A+	A	A-	A	B+	A-	A-	A
SocialPillarScore	B-	B+	B	B	B	B	B+	B+

Table 7.3.16: Landfill Investment

Landfills Investment	2017	2018	2019	2020	2021	2022	2023	2024
Number of landfills owned or operated	195.00	190.00	189.00	186.00	198.00	206.00	207.00	208.00
Net investment, excluding non-depletable land (in millions)	3673.20	3758.00	3872.90	4046.00	4193.30	4873.60	4745.40	4924.00
Total estimated available disposal capacity (in millions of cubic yards)	5086.00	5078.00	4994.70	4988.90	5012.70	5014.30	5104.00	5027.00
Net investment per cubic yard	0.72	0.74	0.78	0.81	0.84	0.97	0.93	0.98
Landfill depletion and amortization expense (in millions)	326.00	325.60	333.00	323.00	376.80	433.70	470.90	514.00
Accretion expense (in millions)	79.80	80.70	81.90	82.90	82.70	89.60	97.90	107.00
	405.80	406.30	414.90	405.90	459.50	523.30	568.80	621.00
Airspace consumed (in millions of cubic yards)	82.00	83.20	81.50	76.10	79.30	85.00	85.90	87.00
Depletion, amortization and accretion expense per cubic yard of airspace consumed	4.95	4.88	5.09	5.33	5.79	6.16	6.62	7.14

Table 7.3.17: Accrued Landfill and Environmental Costs

Accrued Landfill and Environmental Costs	2017	2018	2019	2020	2021	2022	2023	2024
Landfill final capping, closure and post-closure liabilities	1,257.70	1,292.00	1,335.60	1,346.40	1,507.30	1,786.40	1,937.20	2,144.00
Environmental remediation	564.00	540.20	500.20	462.80	454.90	487.50	485.40	447.00
Total accrued landfill and environmental costs	1,821.70	1,832.20	1,835.80	1,809.20	1,962.20	2,273.90	2,422.60	2,591.00
Less: current portion	-135.20	-130.60	-132.60	-114.50	-124.50	-132.60	-141.60	-159.00
Long-term portion	1,686.50	1,701.60	1,703.20	1,694.70	1,837.70	2,141.30	2,281.00	2,432.00

7.4 Combined Data

Figure 7.4.1: Comparative Total Shareholder Return Chart: 10-Year Trends (2014-2024)

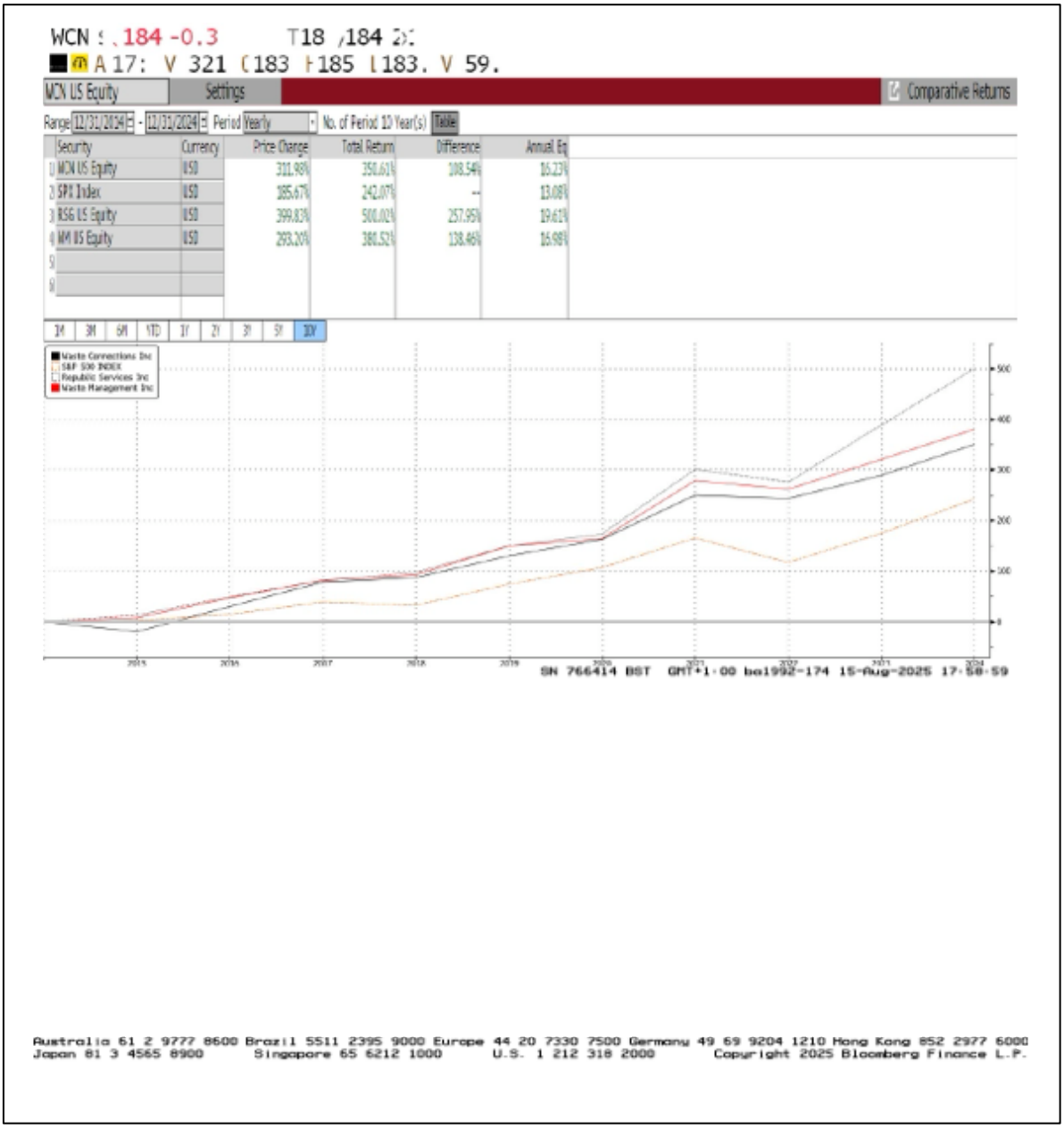


Figure 7.4.2: Comparative Total Shareholder Return Table: 10-Year Trends (2014-2024)



COMP Comparative Returns — WCN US Equity — Options

Range 12/31/2014 - 12/31/2024 Period Yearly No. of Period 10 Year(s) Chart

	WCN US Equity			SPX Index			RSG US Equity			WM US Equity		
Date↓	Price	Divs	Return	Price	Divs	Return	Price	Divs	Return	Price	Divs	Return
12/31/2024	171.58	1.17	350.61%	5881.63	73.80	242.07%	201.18	1.65	500.02%	201.79	3.00	380.52%
12/29/2023	149.27	1.05	289.39%	4769.83	69.38	173.66%	164.91	2.06	387.69%	179.10	2.80	320.45%
12/30/2022	132.56	.95	243.15%	3839.50	66.32	116.74%	128.99	1.91	276.21%	156.88	2.60	261.91%
12/31/2021	136.27	.85	250.26%	4766.18	59.56	164.72%	139.45	1.77	300.89%	166.90	2.30	278.97%
12/31/2020	102.57	.76	161.78%	3756.07	57.94	105.72%	96.30	1.66	172.66%	117.93	2.18	163.51%
12/31/2019	90.79	.67	129.92%	3230.78	57.94	73.77%	89.63	1.56	148.93%	113.96	2.05	149.80%
12/31/2018	74.25	.58	86.63%	2506.85	53.45	32.17%	72.09	1.44	96.62%	88.99	1.86	91.49%
12/29/2017	70.94	.50	76.95%	2673.61	48.66	38.24%	67.61	1.33	80.64%	86.30	1.70	81.83%
12/30/2016	52.39	.51	29.66%	2238.83	45.46	13.48%	57.05	1.24	49.35%	70.91	1.64	46.10%
12/31/2015	32.61	.70	-20.20%	2043.94	43.12	1.37%	43.99	1.16	12.42%	53.37	1.54	7.13%
12/31/2014	41.65	.77	.00%	2058.90	--	.00%	40.25	1.08	.00%	51.32	1.50	.00%

Table 7.4.3: ROE and CE Revenue as a Percentage of Revenue

Company	Year	CE % of Revenue	ROE
Waste Connections	2017	3.49%	9.19%
Waste Management	2017	9.89%	32.38%
Republic Services	2017	1.48%	16.07%
Waste Connections	2018	1.88%	8.47%
Waste Management	2018	8.67%	30.65%
Republic Services	2018	1.94%	13.09%
Waste Connections	2019	1.19%	8.17%
Waste Management	2019	6.73%	23.64%
Republic Services	2019	1.86%	13.22%
Waste Connections	2020	1.59%	2.98%
Waste Management	2020	7.41%	20.08%
Republic Services	2020	1.61%	11.42%
Waste Connections	2021	3.33%	8.84%
Waste Management	2021	9.37%	25.51%
Republic Services	2021	2.15%	14.39%
Waste Connections	2022	2.84%	11.75%
Waste Management	2022	7.70%	32.71%
Republic Services	2022	9.34%	15.36%
Waste Connections	2023	1.83%	9.91%
Waste Management	2023	6.19%	32.97%
Republic Services	2023	11.37%	16.42%
Waste Connections	2024	2.71%	7.86%
Waste Management	2024	7.27%	33.26%
Republic Services	2024	11.89%	17.92%

Table 7.4.4: EBIT Margin

EBIT Margin	2017	2018	2019	2020	2021	2022	2023	2024
Waste Management	18.20%	18.70%	17.51%	15.99%	16.54%	17.08%	17.50%	18.42%
Waste Connections	13.54%	16.91%	15.55%	7.57%	16.90%	17.22%	15.41%	11.97%
Republic Services	16.62%	17.29%	17.35%	16.83%	18.38%	17.70%	18.58%	19.94%

Table 7.4.5: DD&A as a percentage of revenue

DD&A % of revenue	2017	2018	2019	2020	2021	2022	2023	2024
Waste Management	9.50%	9.90%	10.18%	10.98%	11.15%	10.35%	10.14%	10.28%
Waste Connections	13.66%	13.82%	13.81%	13.82%	13.22%	12.74%	12.51%	13.05%
Republic Services	10.32%	10.29%	10.10%	10.60%	10.50%	10.01%	10.03%	10.46%

Table 7.4.6: Total Debt-to-Total Asset Ratio

Total Debt-to-Total Asset	2017	2018	2019	2020	2021	2022	2023	2024
Waste Management	43.48%	44.26%	48.65%	47.06%	46.07%	47.77%	49.44%	53.63%
Waste Connections	32.55%	32.91%	31.70%	33.71%	34.33%	40.25%	37.68%	40.78%
Republic Services	38.72%	38.57%	38.30%	38.12%	38.29%	40.57%	40.81%	39.24%

Table 7.4.7: EV/EBITDA

EV/EBITDA	2017	2018	2019	2020	2021	2022	2023	2024
Waste Management	3.24	3.20	3.80	4.18	4.64	4.02	4.32	4.79
Waste Connections	17.84	15.3	17.53	26.1	21.59	18.82	20.05	23.15
Republic Services	10.96	11.06	12.7	13.68	15.94	13.56	14.6	15.04

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